(10) Factorize each of the following completely:

① $x^2 - y^2 - 6x - 6y$

② $x^2 - a^2 - 2x + 1$

③ $x^4 + 2x^3 - 9x^2 - 18x$

④ $4x^2 - 12xy + 9y^2 - 49$

⑤ $5a^2b - 10a^2b^2 + 4b^2 - 8b^3$

⑥ $m^3 + n^3 - m - n$

⑦ $4k^4 - 49k^2 + 14k - 1$

⑧ $16x^2 - 25a^2 + y^2 + 8xy$

(11) Factorize each of the following completely:

① $x^4 + 4y^4$

② $4m^4 + 1$

③ $64x^4 + y^4$

④ $a^4 + a^2b^4 + b^8$

⑤ $y^4 - 23y^2 + 1$

⑥ $x^4 - 11x^2 + 1$

⑦ $16a^4 + 20a^2b^2 + 9b^4$

⑧ $k^5 + 5k^3 + 9k$

⑨ $4x^2(4x^2 - 7y^2) + y^4$

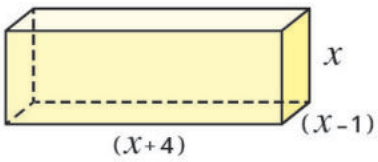
(12) Find the solution set of each of the following equations in R:

① $x^3 + x^2 - 4x = 4$

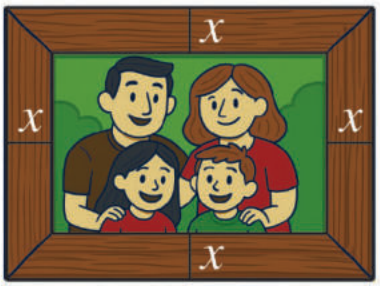
② $x^2 + 12x + 15 = 0$

Third Analysis and Subjects Integration

(13) Geometry: If the volume of the box is 12 cubic units, find its dimensions.



(14) Geometry: If the volume of a cuboid is $(x^4 - 2x^3 + 3x^2 - 6x)$ cubic feet, find possible dimensions of the cuboid in terms of x, then find the numerical values of the dimensions when $x = 4$



(15) Decoration: A picture is surrounded by a frame with a thickness of x inches, as shown in the figure. If the area of the picture is $(4x^3 + 6x^2 + 2x + 3)$ square inches, find possible dimensions of the framed picture in terms of x.

Creative Thinking

(16) If $a + b = 3$, find the value of $a^3 + 3a^2b + 3ab^2 + b^3$

(17) Factorize completely: $x^8 - 16y^8$

***Evaluate** your understanding!*

How well do you understand factorizing by grouping?
Tick the right box